Joseph Gabriel Hernandez

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EDUCATION

Bachelor of Science in Electrical Engineering **Graduated: December 2025**

San Francisco State University, San Francisco, CA

- Concentration in Embedded Systems and Power Systems

Relevant coursework: Design with Microprocessors, Electromechanical Systems, Control Systems, Communications Systems, Electromagnetics, Microelectronics, Digital Signal Processing, Power Systems, Engineering Cost Analysis, Advanced design with microcontrollers, Dynamics, Digital Design, Unix and Linux for Engineering, Power Systems.

PROFESSIONAL EXPERIENCE

Project Engineer Internship **June 2024 - August 2024**

Anvil Power Inc, Emeryville, CA

- Reviewed and revised electrical design plans for substations and transformers, ensuring IEEE and NFPA standards while conducting daily site inspections and system testing.
- Managed RFIs and submittals, optimized procurement processes, and ensured coordination with PG&E for permit approvals.
- Optimized Power Distribution with engineers and project managers to increase UPS and SwitchGear performance, resolving inefficiencies and overlooked details while providing key technical insights in meetings.

Developer &Teacher **May 2022 - Aug 2022**

Inetegem, San Rafael, CA

- Maintained hardware/software integrations, ensuring functionality and reliability for interactive educational tools.
- Taught 15+ students in coding, augmented reality, and programming through interactive lessons to boost interaction and engagement from subjects.

SKILLS AND INTERESTS

- Software & Tools: MATLAB, LTspice, EAGLE, STM32, Keil uVision5, system/substation diagrams, Oscilloscope, Waveform generator, Laboratory Equipment, Soldering, Multimeter, MATLAB, Simulink, Xilinx, ARM/STM32 series, KICAD. Hands-on experience building and debugging hardware, self-motivated,Logic analyzer and scopes,i2C uart, Strong analytical and problem-solving,customer service
- Programming: C, C++, HTML, CSS, JavaScript, Java, VHDL

PROJECTS

Self-Sustaining Greenhouse System - Designed an automated irrigation & temperature control system with ARM-STM32L476RG using Keil uVision5 with assembly and C.

- Programmed with Keil uVision5 to manage led indicators, adc sensors, & timer logic.

Custom Audio Device - Headphones - Designed and built a headphone amplifier using Eagle for schematic design and PCB layout.

- Integrated an OP27 operational amplifier with BJT and MOSFET transistors for optimized audio performance.
- Assembled and tested components, ensuring minimal signal noise and high-quality output.

Metered Power Outlet-

- Designed and implemented a system that powers and measures the data of each socket with remote termination from any location via website/app
- Designed and Ran Hardware validation using VAC LTspice and KiCad for the schematic
- Developed scripts for Hardware validation and testing for hardware.